CP Generator

Generation, Diagnostics, and Animation of Origami Crease Patterns

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What this project is

A desktop app that generates, diagnoses, and animates origami crease patterns on a square sheet — entirely from scratch.

`cp.py` · `cp_gen.py` · `fold_sim.py`

Five linked algorithms

1. Build a planar candidate graph.
2. Repair parity & optimize for Kawasaki.
3. Assign mountains/valleys.
4. Extract faces & animate the fold.
5. Diagnose what is only local, what is globally consistent, and what is still heuristic.

Key objects

- Sheet $[0, s]^2$; graph $G = (V, E)$, positions $X_v \in \mathbb{R}^2$.
- Partial crease labeling $\tau : E \to \{U, M, V\}$.
- Complete MV assignment $\sigma : E \to \{-1, +1\}$, with $+1 = M$ and $-1 = V$.
- Sector angle: angle between consecutive creases at a vertex.
- Half-edge: oriented edge — determines which face lies to its left.

Algorithm I — Planar candidate graph

Goal. A planar graph dense enough for interesting folds, regular enough for stable local angle computation.

Construction

1. Sample interior points; add four corners.
2. Snap near-boundary points to the boundary.
3. Compute the Delaunay triangulation.
4. Use its edges as the initial crease graph.

Why Delaunay?

No other input point lies inside the circumcircle of any triangle — avoids skinny triangles and gives clean local angle data.

Algorithm II — Parity repair & Kawasaki optimization

Necessary condition

Every interior vertex must have even degree: $d(v) \equiv 0 \pmod 2$.

Repair. Pair odd-degree vertices; delete the shortest path between each pair. (Heuristic — not globally optimal.)

Kawasaki's theorem (local)

Sector angles $\alpha_1, \ldots, \alpha_{2m}$ must satisfy

$$\alpha_1 + \alpha_3 + \cdots = \alpha_2 + \alpha_4 + \cdots = \pi.$$

Geometry repair. Let $D(X) = \sum_v w_v \, \|X_v - X_{0,v}\|^2$. Move vertices as little as possible, subject to the Kawasaki constraints:

$$\min_X \; D(X) + \frac{\alpha}{2} D(X)^2,$$

Boundary weights w_v are large; interior weights are small. Solved with SLSQP.

Algorithm III — Local constraints: Maekawa & Bern–Hayes

Maekawa's theorem (local)

At every interior vertex,

$$M(v) - V(v) = \pm 2.$$

Used as an **admissibility filter**: reject any assignment violating this anywhere.

Bern–Hayes-style reduction

For each vertex, repeatedly remove the smallest sector: its two bounding creases are forced *equal* or *opposite* in sign.

1. Sort creases around the vertex by angle.
2. Find the smallest wedge; record its forced sign relation.
3. Recurse until the vertex is fully resolved.

Boundary vertices get temporary virtual rays to close the cyclic order.

Algorithm III — Merging, exact search, & interlacing

Each vertex reduction yields sign chains:

$$\sigma(e_i) = \sigma(e_j) \quad \text{or} \quad \sigma(e_i) = -\sigma(e_j).$$

Exact search

- Merge chains across shared creases into g free groups.
- Enumerate all 2^g seed assignments; propagate implied signs.
- Keep only assignments that pass Maekawa at every vertex.

Undetermined creases keep label -1; no sign is invented.

Interlacing preference

Among admissible assignments, prefer $M, V, M, V, \ldots$ over $M, M, V, V, \ldots$; weight sharper wedges more.

Does not affect correctness.

Algorithm IV — Faces, reflection & 3D animation

Face extraction

Trace half-edges in order; each bounded counterclockwise cycle defines one face. Faces are then triangulated by ear clipping for rendering.

Reflection rule

If two faces share crease ℓ, the child is the parent reflected in ℓ. Propagated from a root face outward; if any cycle is inconsistent, the method falls back to per-triangle propagation along a spanning tree.

Kinematic animation

Around crease axis (p, q), with sign $\sigma(e) \in \{\pm 1\}$ and fold angle θ:

$$T_{\text{child}} = R_{(p,q)}\big(\sigma(e)\,\theta\big)\,T_{\text{parent}}.$$

Rigid plates connected by hinge creases — no elasticity, no collisions. Near the fully folded state, small offsets along the surface normal keep layer order visible.

Algorithm V — Diagnostic status map

Let $\mathcal{C} = (G, X, \tau)$ be the current crease pattern, let $V_{\text{int}} \subseteq V$ denote the interior vertices, and let

$$\kappa(v) = \left| \sum_{i \text{ odd}} \alpha_i(v) - \pi \right|$$

be the Kawasaki residual at v, where $\alpha_1(v), \ldots, \alpha_{2m(v)}(v)$ are the sector angles in cyclic order.

Reported statuses

1. **Local status = pass** iff every $v \in V_{\text{int}}$ has even degree and $\kappa(v) \leq \varepsilon$.
2. **Assignment status = pass** iff τ is complete and $|M_\sigma(v) - V_\sigma(v)| = 2$ for every $v \in V_{\text{int}}$.
3. **Global status = pass** iff the current embedding has no nonincident crease crossings, every crease borders exactly two traced faces, and reflection propagation is cycle-consistent on the face-adjacency graph.
4. **Exact-preview status = pass** iff the current geometry admits exact face reconstruction with no provisional signs and no mesh fallback.

Non-certifying outcomes

A warning is reported whenever exact reconstruction needs a reference-geometry reset, provisional completion of τ, or mesh fallback.

Algorithm VI — Implemented global-consistency test

The current code does *not* solve arbitrary global flat-foldability. It implements the following certifier on the current embedding X:

Implemented decision procedure

1. If nonincident fold segments cross, return `fail`. If there are no folds, return `not_run`. If no MV assignment has been attempted, return `unknown`.
2. Augment the fold graph by the square boundary, and order neighbors at each vertex by polar angle.
3. Trace bounded faces by the left-face rule $(u, v) \mapsto (v, w)$, where w is the predecessor of u in the cyclic order at v. Reject repeated vertices or nonclosing walks.
4. Require every fold edge to border exactly two traced faces. Build the face-adjacency graph, choose a maximal-area root face, and set $T_{f_0} = I$.
5. Across each crease ℓ, propagate $T_g = R_\ell T_f$. On revisiting a face, compare the competing transforms: hard discrepancy gives `fail`, soft discrepancy gives `warning`.
6. Resolve fold signs from τ. Any missing sign is filled provisionally by face-depth parity, which downgrades the outcome to `warning`.

Output interpretation

A final `pass` means: no crossings, successful exact face tracing, two-sided incidence for every fold,

Scope of the local certificates

Theorem 1 (Hull's local flat-foldability conditions)

If a single-vertex crease pattern flat-folds, then its degree is even, its sector angles satisfy Kawasaki's alternating-sum identity, and every complete MV assignment satisfies Maekawa's relation $|M - V| = 2$. [Hul94]

Interpretation in this project.

The implemented *local status* and *assignment status* test these vertexwise necessary conditions, up to numerical tolerance ε. Bern–Hayes reduction is used only to derive local sign constraints and shrink the search over complete assignments σ; it does not decide global flat-foldability for a multi-vertex pattern. Hence a pass certifies *local admissibility*, not a global folding. [Hul94, BH96] □

Why this distinction matters

Hull's theorem is local, while Bern–Hayes shows that exact flat foldability becomes globally difficult once many vertices interact. The report therefore keeps "locally admissible assignment" and "globally certified current geometry" separate. [BH96]

Meaning of a global or exact-preview pass

Theorem 2 (Implemented exact-current-geometry certificate)

If the algorithm on the previous slide returns global status = pass and exact-preview status = pass, then the current embedding of $\mathcal{C}$ has a noncrossing augmented planar graph, a closed family of bounded traced faces, a two-face incidence relation on every fold edge, and a cycle-consistent family of flat reflections $T_g = R_\ell T_f$, all without provisional completion of the MV data.

Why this theorem is faithful to the code.

1. ‘cp.py‘ first rejects nonincident segment crossings.
2. ‘fold_sim.py‘ then traces bounded faces by cyclic half-edge traversal and aborts on repetition or nonclosure.
3. The exact solver requires exactly two incident faces per fold and propagates transforms by explicit crease reflections.
4. Cycle drift beyond tolerance aborts; smaller drift or provisional sign completion downgrades to warning rather than pass.

Thus the theorem states exactly what the present code certifies. It is not yet a complete decision procedure for arbitrary multi-vertex flat-foldability, but it is consistent with the formal viewpoint that valid flat foldings

Certified today vs. unresolved

Currently certifiable

- Vertexwise even-degree and Kawasaki residual checks.
- Maekawa verification on a complete assignment σ.
- Explicit detection of nonincident crease crossings.
- Exact face reconstruction on the current geometry.

Still non-certifying

- Parity repair is not globally optimal.
- Bern–Hayes reduction is local, not a full global classifier.
- Reference-geometry and mesh preview fallbacks remain heuristic.
- A true exact global checker is still needed for larger patterns.

Best theoretical next step

The most plausible upgrade path is a bounded-instance exact checker based on ply and cell-adjacency complexity, built on the face arrangement already computed by the exact preview solver. [Epp23]

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